FAME: Fusion of Alignment and Multiview Enhancement for Remote Sensing Image–Text Retrieval

Yuanhao Su, Daoye Zhu, Zhande Dong, Qifeng Lin, *Member, IEEE*, Lingbo Liu, and Shuming Bao

***Abstract*—Contemporary advancements in Earth observation technologies have generated substantial data resources for remote sensing image retrieval applications. However, existing models exhibit limitations in extracting local features from image–text pairs and establishing effective connections between local and global information. Furthermore, these models lack fine-grained alignment capabilities between different modalities. To address these challenges, we propose FAME, which integrates four key components: a progressive masking (ProgMask) module, an OmniView fusion (OVF) module, a global feature attention with multiview feature alignment (GFA-MFA) module, and regional clustering alignment loss (RCA-Loss). The ProgMask module employs a progressive training strategy with alternating masking of image and text modalities to enhance cross-modal learning robustness. The OVF module employs omni-view attention mechanisms with adaptive weighting to dynamically focus on different local semantic regions of remote sensing images and corresponding textual descriptions. The GFA-MFA module enhances global feature capture through selective filtering while ensuring precise multiview feature alignment. These modules work synergistically to achieve fine-grained alignment between multiscale local and global features across modalities. In addition, RCA-Loss optimizes intraclass feature clustering while minimizing interclass confusion through class center distance optimization and enhanced feature discrimination. Experimental validation on three benchmark datasets (RSITMD, RSICD, and UCM-captions) demonstrates that FAME achieves superior performance compared with existing state-of-the-art methods in remote sensing image–text retrieval (RSITR) tasks.**

***Index Terms*—Cross-attention mechanism, feature alignment, image–text retrieval, local–global information, remote sensing.**

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I. Introduction

REMOTE sensing image–text retrieval (RSITR) is a cross-modal retrieval task that aims to establish semantic bridges between textual descriptions and images to enable efficient retrieval of relevant information from large-scale geospatial datasets [1], [2], [3]. With the rapid growth of remote sensing data volumes, developing automated systems capable of accurately retrieving images based on textual descriptions or vice versa has become increasingly important. These systems enhance the applicability of remote sensing data across multiple domains, including environmental monitoring, urban planning, disaster response, and agricultural management [4], [5], [6], [7], and marine remote sensing applications. Notably, underwater remote sensing has emerged as an extension of traditional remote sensing, where cross-modal retrieval technologies have been successfully applied to underwater image captioning and scene understanding [8], [9]. Remote sensing images exhibit high diversity and complexity, containing rich spatial information and diverse geographical features. However, textual descriptions of remote sensing images are typically brief and cannot comprehensively and accurately reflect all details and information within the images. Multiple images may correspond to a single textual description. Therefore, bridging the semantic gap between images and text to ensure accurate mapping of semantic information extracted from textual queries to image features remains a significant challenge.

Recently, researchers have made significant advances in RSITR tasks, developing numerous retrieval models. Early approaches typically performed global feature extraction on remote sensing images and text, subsequently reducing the distance between image–text features to achieve alignment [10], [11], [12]. However, these methods fail to effectively capture local features in remote sensing image–text pairs, limiting their ability to acquire comprehensive semantic knowledge. Current models typically extract local features and fuse them with global features. Yang et al. [13] proposed a multiscale alignment method that combines single-scale image features with local text features through cross attention. Chen et al. [14] developed the context-aware local–global semantic alignment (CLGSA) method, which improves image–text alignment accuracy through local image occlusion reconstruction and global semantic alignment. Zhang et al. [15] introduced a region-level image–text alignment optimization approach that maps image and text features into the same embedding space, enhancing model generalization capability. Zhang et al. [16]

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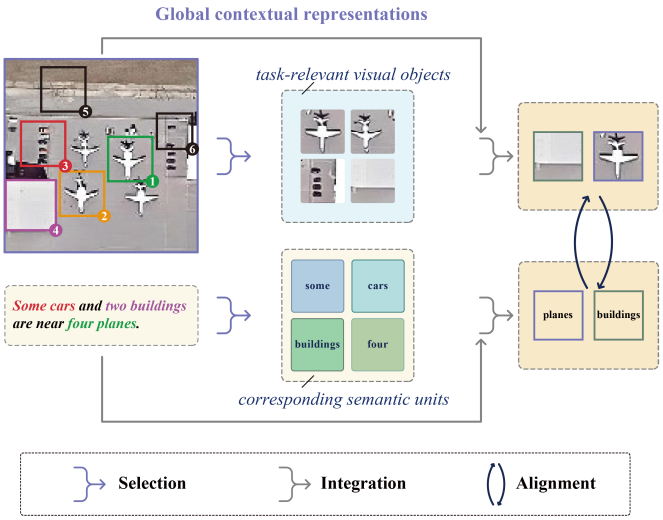


Fig. 1. Three-stage theoretical paradigm of FAME: selection → integration → alignment. The framework establishes a systematic approach through three fundamental stages. 1) *Selection stage* identifies and extracts salient features from both image and text modalities, filtering task-relevant visual objects (highlighted by colored bounding boxes) and corresponding semantic units from textual descriptions to focus computational resources on discriminative information. 2) *Integration stage* synthesizes the selected salient features with global contextual representations through omni-perspective fusion mechanisms, creating comprehensive feature representations that capture both local details and global semantics across modalities. 3) *Alignment stage* performs cross-modal feature interaction and alignment through bidirectional learning strategies, establishing precise correspondences between integrated image and text features to achieve fine-grained semantic matching.

proposed the iEBAKER method, which employs an eliminate before align (EBA) strategy to remove weakly correlated sample pairs, ensuring accurate local feature alignment and utilizes a symmetric alignment retrieval (SAR) strategy to improve global semantic alignment. Yuan et al. [17] achieved bidirectional enhancement by using local features to correct global features while supplementing local information with global features. Sun et al. [18] designed Swin Transformer blocks that combine local window self-attention with global–local window cross attention to improve multiscale feature capture capability. Nevertheless, despite these advances, existing methods face two fundamental limitations: insufficient synergistic integration between local and global features, particularly when processing complex remote sensing scenes with diverse target scales, and reliance on static training paradigms that cannot optimize feature space organization for improved clustering and discrimination.

To address these limitations, this study proposes an innovative training framework that consists of four key components. The ProgMask module implements progressive masking that differs from random masking in Transformers [19] by employing gradually increasing masking rates and alternating cross-modal guidance, where masked images learn from complete text and masked text learns from complete images. This framework then follows a paradigm of Selection, Integration, and Alignment to achieve precise cross-modal matching, as illustrated in Fig. 1. First, the OVF module employs learnable perspective projection matrices to decompose features into multiple semantic views, moving beyond uniform attention mechanisms [20] that treat all regions equally. Building on this multiperspective foundation, the GFA module performs the Selection Stage, using selective attention filtering to focus on the most relevant features rather than the uniform attention distribution in conventional cross-attention mechanisms [21]. Subsequently, the MFA module handles the Integration Stage, synergistically fusing the features from OVF with the refined global features from GFA. Finally, the Alignment Stage uses RCA-Loss to explicitly pull same-class features toward class centers while pushing different-class features beyond a predefined radius threshold, providing explicit geometric constraints for feature organization, unlike contrastive learning [22] that relies solely on implicit pairwise relationships. This integrated approach ensures that each component contributes to a cohesive framework where multiperspective local features and refined global features work synergistically to achieve precise cross-modal alignment.

The main contributions of this article are as follows.

1) A comprehensive FAME framework that integrates a progressive training approach with omni-view feature learning and multiscale feature fusion. The ProgMask module implements alternating modality masking with increasing rates, while the OVF module uses learnable projection matrices for comprehensive perspective understanding. The GFA-MFA modules employ selective attention and cross-modal feature alignment to capture both global context and fine-grained details, collectively enhancing cross-modal robustness and establishing precise correspondences in remote sensing imagery.
2) RCA-Loss that clusters same-category image–text pairs while separating different categories through radius-controlled feature space organization. This loss reduces intercategory interference and promotes more discriminative representations.
3) Extensive experiments across three public datasets (RSITMD, RSICD, and UCM-captions) demonstrate consistent improvements over state-of-the-art methods, confirming the effectiveness of our integrated approach for RSITR.

The remainder of this article is organized as follows. Section II introduces related methods for RSITR tasks. In Section III, we present how our proposed method addresses existing challenges and improvements. Parameter analysis, ablation studies, and comparative experiments on three public datasets are presented in Section IV. Finally, Section V provides conclusion and future direction.

II. Related Works

This section provides a comprehensive review of RSITR literature through the following two approaches: 1) related methods for RSITR and 2) cross-attention mechanisms.

A. Related Methods for RSITR

Existing RSITR models can generally be categorized into two main types: methods based on traditional dual-stream architectures and methods based on pretraining and fine-tuning paradigms.

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1) Methods Based on Traditional Dual-Stream Models: Traditional dual-stream image–text retrieval models employ separate image and text encoders to process visual and textual information, mapping them into the same semantic space for similarity computation. Contrastive losses such as triplet loss [23] and contrastive loss [22] ensure that images and corresponding texts are positioned closely in the shared feature space while pushing apart unrelated pairs.

Early improvements to dual-stream architectures focused on optimizing embedding spaces and pooling strategies. Faghri et al. [24] proposed VSE++ with hard negative mining and improved contrastive loss, while Chen et al. [25] introduced VSE∞ with generalized pooling operators for adaptive feature processing.

For remote sensing applications, several domain-specific methods have been developed to address unique challenges. Yuan et al. [26] proposed AMFMN for handling multiscale scarcity and positive sample ambiguity, while Pan et al. [27] developed PIR to solve semantic noise problems using prior knowledge and self-attention encoders. Other notable contributions include ROAM [28] for adaptive embedding adjustment, MTGFE [29] for multiview representation learning, and VMSF/TCGE [13] for addressing semantic confusion through multiscale fusion and context-aware generation.

To achieve finer grained integration between images and text, researchers have explored global and local information alignment approaches. For general image–text retrieval, representative methods include CAMP [30] for fine-grained relationship exploration, SCAN [21] for cross-attention alignment, and VisualBERT [31] for automatic attention tracking. In the remote sensing domain, Yuan et al. [17] developed the GLAR framework with multilevel dynamic fusion and denoising mechanisms, while Zhao et al. [32] introduced GLISA/ALIE modules for semantic ambiguity alleviation. Yang et al. [13] proposed MSA methods based on MSCMAT to address multiscale semantic alignment through cross-modal and cross-scale consistency losses.

Despite these advances, traditional dual-stream models often struggle with noise interference and fine-grained image–text processing, resulting in suboptimal accuracy for RSITR tasks.

2) Methods Based on Pretraining and Fine-Tuning: In RSITR tasks, methods based on pretraining and fine-tuning have become prevalent and effective approaches. These methods leverage general features and knowledge learned from large-scale datasets to help models better adapt to remote sensing image–text matching tasks. To address the computational resource challenges of fully fine-tuning vision–language pretraining (VLP) models and the issues arising from continuously updated remote sensing data, Yuan et al. [33] proposed PE-RSTIR, a parameter-efficient transfer learning method based on the CLIP model [34], which efficiently transfers vision–language knowledge from the natural language domain to the remote sensing domain. Huang et al. [35] proposed HarMA, a method that harmonizes transfer learning with modality alignment, minimizing training overhead through coordinated task constraints, modality alignment, and unimodal uniform alignment.

Several CLIP-based improvement methods have also been developed. Mo et al. [36] proposed S-CLIP, which significantly improves performance with limited image–text pairs by introducing caption-level and keyword-level pseudo-labels. Addressing similar challenges, Liu et al. [37] proposed RemoteCLIP, which created a large-scale dataset through box-to-caption and mask-to-box methods, achieving significant performance improvements after pretraining on this dataset. Zhang et al. [38] proposed GeoRSCLIP while creating the RS5M dataset, resulting in improved model performance after fine-tuning. Combining modality-level distribution consistency regularization loss with self-pruning distillation strategies, Yang et al. [39] proposed the AIR framework. This framework addresses intramodal distribution consistency issues within the retrieval space while reducing CLIP's computational costs. Liao et al. [40] contributed the explicit key local information selection and reconstruction (EKLSR) framework, which leverages hidden knowledge in pretrained models to select and fuse key local information while enhancing cross-modal and intramodal semantic alignment. Based on BLIP [41], Mi et al. [42] proposed KTIR, which utilizes information from external knowledge graphs to expand query text content, addressing the information asymmetry between text and images.

Although existing RSITR models have achieved numerous advances in this field [1], [43], limitations still exist when processing image–text data. These models often fail to effectively handle noise interference in images and text and struggle with fine-grained image–text processing, resulting in suboptimal accuracy in RSITR tasks.

B. Cross-Attention Mechanisms

Currently, an increasing number of researchers are incorporating attention mechanisms into remote sensing vision–language models. The attention mechanism was first proposed by Bahdanau et al. [44] for addressing machine translation problems. Subsequently, Vaswani et al. [20] introduced the Transformer architecture, a neural network architecture entirely based on attention mechanisms. To effectively fuse different types of information, cross-attention mechanisms have emerged as a natural solution.

Hou et al. [45] introduced cross-attention modules to construct attention maps that emphasize the most relevant category features. To better utilize contextual information in images, Huang et al. [46] proposed CCNet, which enhances global information capture capabilities in images. Wei et al. [47] proposed the MMCA network. Unlike traditional approaches, this model performs modeling not only within image and text modalities but also facilitates information transmission between modalities. Consequently, both intramodal and intermodal image–text features are enhanced, improving the accuracy of image-sentence matching.

Building upon ViT [48], Chen et al. [49] proposed CrossViT, which employs a dual-branch structure to process image patches of different sizes. These patches undergo multiple cross-fusion operations to complement each other's information. Compared with traditional attention methods, this model achieves linear computational and memory complexity.

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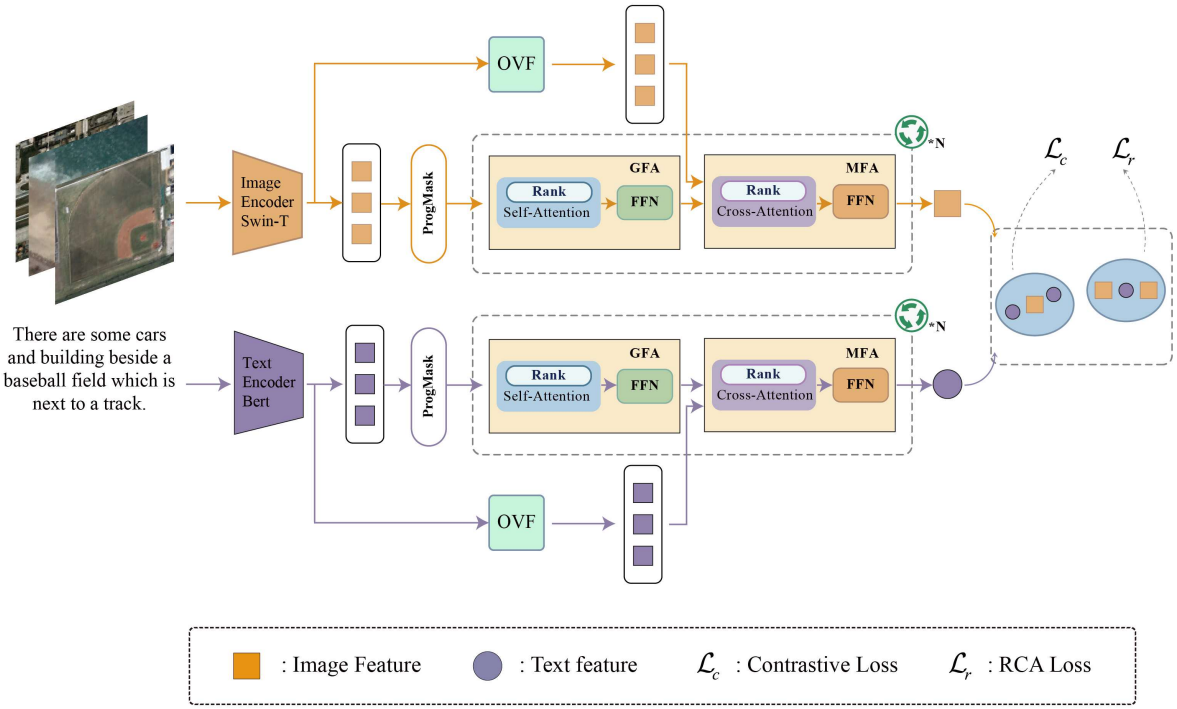


Fig. 2. Overall architecture of the FAME framework for RSITR. The framework consists of five main components: 1) feature extraction using Swin Transformer and BERT; 2) ProgMask module for progressive masking strategy; 3) OVF module for omni-perspective feature extraction; 4) GFA-MFA module for iterative cross-modal feature fusion; and 5) loss optimization using contrastive loss ($\mathcal{L}_c$) and regional clustering alignment loss ($\mathcal{L}_r$).

To address the computational overhead of traditional Transformers in computer vision tasks, Lin et al. [50] proposed a hierarchical cross-attention network. This network reduces computational costs by alternately executing local and global attention while enhancing feature fusion capabilities between different image regions.

To address the fusion of different modal features in traditional images, Shen et al. [51] proposed a bidirectional cross-attention transformer that aggregates features from two modalities from both local and global perspectives to better capture their complementary characteristics. Addressing similar challenges, Li and Wu [52] utilized cross-attention modules (CAMs) to enhance complementary information while reducing redundant features.

In summary, cross-attention mechanisms play an increasingly important role in multimodal feature fusion for image–text tasks, significantly improving the performance of image–text understanding tasks. In RSITR tasks, cross-attention mechanisms can establish local and global relationships between remote sensing images and textual descriptions, thereby improving the accuracy of remote sensing image–text encoding.

III. Methods

This section introduces the FAME model proposed for the RSITR task. Fig. 2 illustrates the overall framework of the model. The model employs Swin Transformer [53] as the visual encoder and BERT [19] as the text encoder. The model comprises several key components, including the ProgMask module, OVF module, GFA-MFA module, and RCA-Loss. ProgMask implements a progressive masking strategy for enhanced cross-modal learning. Detailed discussions of these modules are presented in Sections III-A–III-D.

A. Overall Framework

As illustrated in Fig. 2, for RSITR tasks, this study integrates Swin Transformer and BERT models to effectively process image and text information. In the visual encoding module, Swin Transformer employs a window-based mechanism to capture global image features. Given an input image $I \in \mathbb{R}^{H\times W\times C}$, where H denotes the image height, W represents the width, and C indicates the number of channels, the image is partitioned into multiple fixed-size windows. Each window extracts local features through a self-attention mechanism. Through sliding window operations, information is exchanged across the entire image scope, ultimately obtaining global features. In the text encoding module, BERT utilizes a bidirectional Transformer architecture to encode input text $T = \{t_1, t_2, \ldots, t_m\}$, transforming it into word vectors. The multilayer Transformer encoder subsequently generates global semantic representations of the text. Through this approach, the model achieves effective information mapping and fusion between images and text, establishing a solid foundation for further optimization of RSITR tasks.

After obtaining global features from both images and text, we input them into the OVF module to extract omni-perspective features of images and text, thereby enhancing feature diversity and representational capability. Subsequently, through the cyclic GFA-MFA module, multilayered fusion is performed on different features of the same modality, enabling

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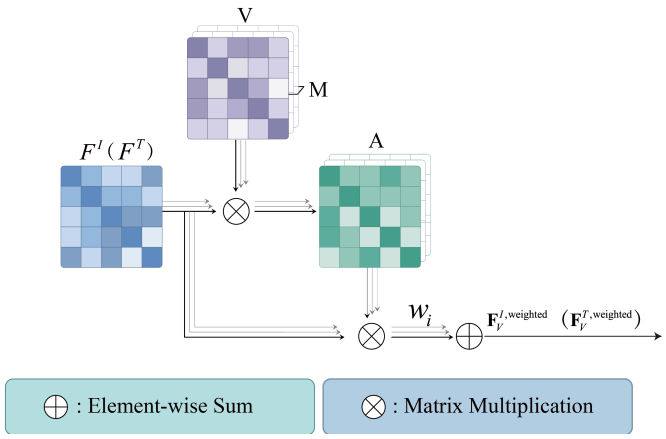


Fig. 3. OVF module architecture. The module employs M learnable perspective projection matrices to decompose input features into multiple semantic views, computing attention correlations and performing adaptive weighted fusion to generate enhanced omni-perspective representations for comprehensive feature understanding.

iterative information transfer to more accurately capture complex associations between images and text. Finally, the output image and text features are fed into a similarity distance matrix, where contrastive loss and RCA-Loss are employed for image–text alignment to optimize the model's retrieval performance.

In addition, to enhance the model's generalization ability and robustness to incomplete information, we propose the ProgMask module. This module implements a progressive masking strategy that strengthens cross-modal learning by encouraging the model to infer complete semantic representations from partial information. The ProgMask module operates with an initial masking rate of 0.1, progressively increasing to 0.3 as training advances. During odd-numbered training epochs, the module applies strategic masking to image patches while preserving text information, compelling the model to leverage textual semantics for robust visual feature learning. Conversely, during even-numbered epochs, text tokens are selectively masked while image information remains intact, encouraging the model to utilize visual cues for enhanced textual understanding. The masking rate incrementally increases after every few thousand training iterations, creating a curriculum learning paradigm that gradually increases task complexity. Importantly, this progressive masking approach does not compromise image–text correspondence but rather strengthens it by forcing the model to discover and exploit deeper cross-modal semantic associations. By learning to reconstruct complete multimodal representations from partially masked inputs, the model develops more robust feature encodings that better capture the intrinsic relationships between visual and textual elements, ultimately improving retrieval performance on complete, unmasked data during inference.

B. OmniView Fusion

Although Swin Transformer can extract global semantic information from remote sensing images, these images often contain numerous small targets, causing Swin Transformer to frequently tend to overlook these fine-grained objects during information extraction [54], thereby lacking a comprehensive representation of remote sensing images. Similarly, text faces analogous challenges. In RSITR tasks, sentence lengths are typically short, making it difficult to comprehensively describe every detail of an image [42]. To address this issue, we propose the OVF module, as illustrated in Fig. 3. This module is based on the omni-perspective feature decomposition theory and effectively enhances the perception capability for fine-grained features by constructing complementary semantic projection spaces.

Remote Sensing Image Omni-Perspective Feature Decomposition: Given visual features $\mathbf{F}^I = \{f_1^I, f_2^I, \ldots, f_L^I\} \in \mathbb{R}^{L\times D}$ encoded by Swin Transformer, where L denotes the feature sequence length and D represents the feature dimension, we design M learnable perspective projection matrices $\mathbf{V} = \{V_1, V_2, \ldots, V_{\mathrm{M}}\} \in \mathbb{R}^{D\times L}$. These matrices learn different feature patterns through end-to-end training and self-adaptive learning. Based on the omni-perspective attention mechanism, we establish feature-perspective correlation mapping

$$A_i = \text{SoftMax}\left(\mathbf{F}^I \cdot V_i\right) \tag{1}$$

where this mechanism learns the semantic correlation between perspective V_i and input features $\mathbf{F}^I$, enabling adaptive attention to different semantic regions. Subsequently, perspective-specific representations are obtained through attention reweighting

$$F_i' = \left(\mathbf{F}^I\right)^{\mathrm{T}} \cdot A_i. \tag{2}$$

This process essentially achieves multidimensional decomposition of feature space, enabling the model to understand image semantics from complementary perspectives. To integrate omni-perspective detailed information, we employ an adaptive weighted fusion strategy

$$\mathbf{F}_V^{(I,\text{weighted})} = \sum_{i=1}^{\mathrm{M}} w_i F_i' \tag{3}$$

where the weights w_i are learned through training, ensuring that important perspectives receive greater contribution.

Text Omni-Perspective Feature Enhancement: Similar to image processing, we input text features $\mathbf{F}^{\mathrm{T}} = \{f_1^{\mathrm{T}}, f_2^{\mathrm{T}}, \ldots, f_S^{\mathrm{T}}\} \in \mathbb{R}^{S\times D}$ encoded by BERT into the OVF module to obtain enhanced text representation $\mathbf{F}_V^{(\mathrm{T},\text{weighted})}$, where S denotes the text sequence length and D represents the feature dimension.

C. GFA-MFA

Remote sensing images contain diverse targets with multiple categories, varying shapes, and different sizes. For instance, in the COCO dataset, target sizes are strictly defined as small targets with areas typically less than or equal to 32 pixels [55]. The size disparities among targets in remote sensing images are even more pronounced, with many small targets potentially falling below this threshold, thus requiring special attention [56]. Traditional global feature extraction methods are difficult to comprehensively cover all targets in an image,

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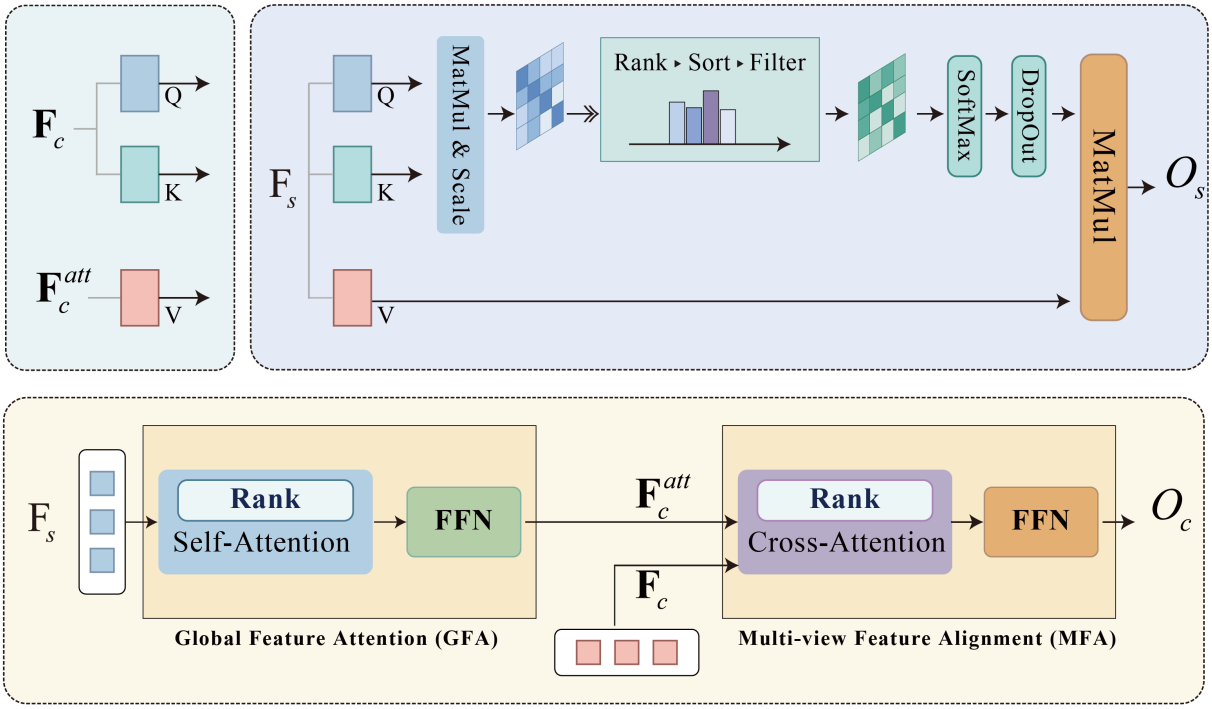


Fig. 4. GFA-MFA module architecture. The module consists of global feature attention (GFA) with top-k filtering and multiview feature alignment (MFA) that fuses GFA and OVF outputs through cross attention for comprehensive feature integration.

causing single global features to inadequately capture the characteristics of each target. Although multihead attention mechanisms compute correlations between different regions through self-attention and enhance model representational capability [20], they often uniformly attend to different image regions when processing remote sensing images with diverse target categories and significant size variations, failing to provide targeted attention to smaller or marginal targets [57]. These targets typically occupy smaller regions or are located at edges, making them prone to being overlooked or difficult to extract [58]. Therefore, although multihead attention mechanisms possess the capability to process local information, their performance in handling small targets remains insufficient in the complex scenario of remote sensing images.

Based on this observation, we design the GFA-MFA module. As illustrated in Fig. 4, we use $\mathbf{F}_s$ as input ($\mathbf{F}_s \in \mathbb{R}^{N\times D}$), which is projected through three linear layers to query $\mathbf{Q} \in \mathbb{R}^{N\times D_k}$, key $\mathrm{K} \in \mathbb{R}^{N\times D_k}$, and value $\mathbf{V} \in \mathbb{R}^{N\times D_k}$ spaces ($D_k = D/h$), as shown in the following equations:

$$\mathbf{F}_s = \left[\mathbf{F}^{\mathrm{T}}, \mathbf{F}^{I}\right] \tag{4}$$

$$Q_i = \mathbf{F}_s \cdot W_i^{Q}, \quad K_i = \mathbf{F}_s \cdot W_i^{\mathrm{K}}, \quad V_i = \mathbf{F}_s \cdot W_i^{V} \tag{5}$$

where the linear projection matrices $W^{Q} \in \mathbb{R}^{D\times D_k}$, $W^{\mathrm{K}} \in \mathbb{R}^{D\times D_k}$, and $W^{V} \in \mathbb{R}^{D\times D_k}$.

Based on query $\mathbf{Q}$ and key $\mathbf{K}$, we perform importance evaluation on each part of the input features and convert them into a score matrix scores $\in \mathbb{R}^{N\times N}$

$$\mathrm{scores}_i = \frac{Q_i \cdot K_i^{\mathrm{T}}}{\sqrt{d_k}}. \tag{6}$$

To more effectively focus on fine-grained and marginal targets in image–text pairs while reducing the noise interference of image–text features, we rank the score matrix and construct a mask topk_mask$_i$. For each query, we retain the top-k keys with the highest scores and set other scores to negative infinity

$$S_i^{(k)} = \text{Top-k}\,(S_i) \tag{7}$$

$$I_i^{(k)} = \text{Indices of the Top-k values from } S_i \tag{8}$$

$$\text{topk_mask}_i = \begin{cases} 1, & \text{if } i \in I_i^{(k)} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

$$\text{scores}_i = \begin{cases} \text{scores}_i, & \text{if topk_mask}_i = 1 \\ -\infty, & \text{if topk_mask}_i = 0. \end{cases} \tag{10}$$

After mask processing, the non-top-k portions in the score matrix are masked, ensuring that the model only focuses on the most relevant information

$$\text{att_map}_i = \text{SoftMax}\,(\text{scores}_i) \tag{11}$$

$$\text{head}_i = \text{att_map}_i \cdot V_i \tag{12}$$

$$\text{MHA}\,(\mathbf{Q}, \mathrm{K}, \mathbf{V}) = [\text{head}_1, \text{head}_2, \ldots, \text{head}_h]^{\mathrm{T}}. \tag{13}$$

Finally, we obtain the weighted output $\mathbf{O}_s$. To enhance the model's representational capability and improve model flexibility, we input the attention matrix into a feedforward neural network

$$\mathbf{O}_s = \text{FFN}\,(\text{MHA}\,(\mathbf{F}_s, \mathbf{F}_s, \mathbf{F}_s)). \tag{14}$$

To effectively process omni-perspective image features and capture correlations between different perspectives as well as correlations between features from different sources, we design

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the MFA module, enabling features from the same modality but different sources to undergo deep cross interaction and mutual complementation. Simultaneously, this approach constructs richer image representations, allowing the model to more comprehensively understand the structure and semantics of image–text pairs, thereby improving task accuracy and robustness. Specifically, we take the original image features and original text features $\mathbf{F}_V^{I,\text{weighted}}$ and $\mathbf{F}_V^{\text{T,weighted}}$ after the OVF module as $\mathbf{F}_c$, and take the original image features $\mathbf{O}_s^I$ and original text features $\mathbf{O}_s^{\text{T}}$ after the self-attention mechanism as $\mathbf{F}_c^{\text{att}}$. Unlike the self-attention mechanism, MFA uses $\mathbf{F}_c$ as query and key, with $\mathbf{F}_c^{\text{att}}$ as value

$$\mathbf{F}_c^{\text{att}} = \left[\mathbf{O}_s^I, \mathbf{O}_s^{\text{T}}\right] \tag{15}$$

$$\mathbf{F}_c = \left[\mathbf{F}_V^{I,\text{weighted}}, \mathbf{F}_V^{\text{T,weighted}}\right] \tag{16}$$

$$\mathbf{O}_c = \text{FFN}\left(\text{MHA}\left(\mathbf{F}_c^{\text{att}}, \mathbf{F}_c, \mathbf{F}_c\right)\right). \tag{17}$$

D. RCA-Loss

To optimize the similarity between images and text, we adopt a contrastive loss approach. Unlike traditional triplet loss, contrastive loss does not require additional negative samples, thereby simplifying the model training process and improving computational efficiency [27]. In our model, features are first extracted from a batch of image–text pairs. Each image I_i and text T_i is processed through embedding networks to obtain their vector representations $\mathbf{v}_i$ and $\mathbf{t}_i$. Subsequently, we compute the cosine similarity $s_{i,j} = \mathbf{v}_i^{\text{T}}\mathbf{t}_j$ between each image and text pair. Based on this, we can derive image-to-text and text-to-image contrastive losses $\mathcal{L}_{v2t}(\mathbf{v}_i, \mathbf{t}_i)$ and $\mathcal{L}_{t2v}(\mathbf{t}_i, \mathbf{v}_i)$. Consequently, we obtain the final contrastive loss $\mathcal{L}_c$

$$\mathcal{L}_{v2t}\left(\mathbf{v}_i, \mathbf{t}_i\right) = -\log\left(\frac{\exp\left(s_{i,i}/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{i,j}/\tau\right)}\right) \tag{18}$$

$$\mathcal{L}_{t2v}\left(\mathbf{t}_i, \mathbf{v}_i\right) = -\log\left(\frac{\exp\left(s_{i,i}/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{j,i}/\tau\right)}\right) \tag{19}$$

$$\mathcal{L}_c = \frac{1}{2}\sum_{i=1}^{N}\left(\mathcal{L}_{v2t}\left(\mathbf{v}_i, \mathbf{t}_i\right) + \mathcal{L}_{t2v}\left(\mathbf{t}_i, \mathbf{v}_i\right)\right) \tag{20}$$

where τ represents the temperature parameter.

Currently, most remote sensing image–text tasks only consider contrastive loss and triplet loss. However, these losses rely on positive sample pairs for optimization and do not focus on the impact of noise and background interference between sample pairs. Meanwhile, these losses fail to address the issue of unbalanced distribution of similar and dissimilar samples in feature space. Based on this observation, we propose RCA-Loss, as illustrated in Fig. 5. Through intraregion class-level loss combined with radius constraints, we reduce noise interference. By aggregating similar features and separating different class features, the model can focus more on key regions, thereby effectively reducing background noise interference. Simultaneously, we introduce a loss term that aggregates similar features while separating dissimilar features, bringing similar features together and pulling apart dissimilar features to ensure clearer class-level distinctions.

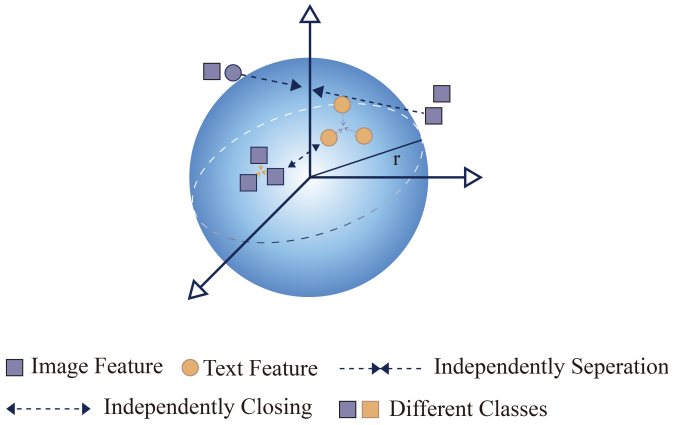


Fig. 5. RCA-Loss framework. The loss function employs radius-controlled clustering to aggregate same-class features within radius r while separating different-class features beyond the radius threshold, achieving balanced feature distribution and reduced noise interference through intraclass aggregation and interclass separation mechanisms. Purple squares represent image features, orange circles represent text features, dashed arrows indicate independent separation (pushing different classes apart) and independent closing (pulling the same classes together), with different colored squares denoting distinct semantic classes.

1) Intraclass Feature Aggregation Loss $\mathcal{L}_{\text{Intra}}$: Features of the same class for both images and text should be clustered together. Therefore, we design an intraclass aggregation loss that optimizes by computing the distance between the same-class image and text features and their class center. Specifically, C_i represents the class center of the ith class and $\mathbf{I}_j$ and $\mathbf{T}_j$ denote the feature vectors of images and text, respectively. We minimize the distance between image and text features and their class center, promoting the aggregation of same-class image and text features, as shown in the following equations:

$$C_i = \frac{1}{|S_i|}\sum_{j\in S_i}\left(\mathbf{I}_j + \mathbf{T}_j\right) \tag{21}$$

$$\mathcal{L}_{\text{intra}} = \sum_{i\in C}\left(\left|\left|\mathbf{I}_j - C_i\right|\right|_2^2 + \left|\left|\mathbf{T}_j - C_i\right|\right|_2^2\right). \tag{22}$$

2) Interclass Feature Separation Loss $\mathcal{L}_{\text{Inter}}$: To ensure that features of different classes for images and text are effectively separated, we design an interclass separation loss. This loss ensures that the distance between features of different classes for images and text is greater than a predefined radius r , effectively reducing feature confusion between different classes, as shown in the following equation:

$$\mathcal{L}_{\text{inter}} = \sum_{i\neq j}\left(\max\left(0, r - \left|\left|\mathbf{I}_i - \mathbf{T}_j\right|\right|_2\right) + \max\left(0, r - \left|\left|\mathbf{T}_i - \mathbf{I}_j\right|\right|_2\right)\right). \tag{23}$$

3) Total Loss: Finally, we combine the intraclass aggregation loss and interclass aggregation loss to obtain $\mathcal{L}_{\text{RCA}}$

$$\mathcal{L}_{\text{RCA}} = \mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}. \tag{24}$$

The final combined loss $\mathcal{L}_{\text{final}}$ is formulated as

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{\text{RCA}} + \mathcal{L}_c. \tag{25}$$

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IV. Experiments

We conducted extensive experiments on the RSITMD, RSICD, and UCM-captions datasets. The experimental results demonstrate that our proposed FAME model is effective for RSITR tasks.

A. Datasets

RSITMD is a fine-grained remote sensing dataset proposed by Yuan et al. [26]. This dataset contains 4743 images with a resolution of 256 × 256 pixels. Each image is accompanied by five descriptive sentences.

RSICD [59] is collected from platforms such as Baidu Maps and Google Earth. It comprises 10 921 remote sensing images with a resolution of 224 × 224 pixels. Each image includes five descriptive sentences.

UCM-captions [60] is a remote sensing image–text dataset proposed by Chinese Academy of Sciences. It contains 2100 images with a resolution of 256 × 256 pixels. Each image is provided with five descriptive sentences.

Following the dataset partitioning methodology from previous studies [17], we uniformly divide the datasets into 70% for training, 20% for testing, and 10% for validation.

B. Evaluation Metrics

Following previous RSITR tasks, we employ two evaluation metrics: R@K (K = 1,5,10) and mR. R@K represents the proportion of relevant results appearing within the top-K retrieval results. mR denotes the mean rank position of relevant images across all queries. A higher ranking indicates better retrieval quality.

C. Implementation Details

All experiments are conducted on an NVIDIA GeForce RTX 4090 GPU, which provides powerful performance for efficient model training. Input images are resized to a final resolution of 256 × 256 pixels. We employ the tiny version of Swin Transformer as the visual encoder with an output dimension set to 512. In addition, we use the base version of BERT as the text encoder, also configured with an output dimension of 512. The model is trained for 100 epochs using the Adam optimizer. We set the batch size to 256, learning rate to 5×10^{-5}, weight decay to 0.01, and dropout rate to 0.2 to mitigate overfitting and enhance model generalization capability.

D. Methods Comparison

In this section, we conduct a detailed comparison of our model's performance against current mainstream and state-of-the-art models, including VSE++ [24], SCAN [21], AMFMN [26], LW-MCR [61], GaLR [17], ACF [62], MGRM-EL [63], SMLGN [64], DOVE [28], MSITA [65], CDMAN [66], MSA [13], and SIRS [67], on the RSITR task.

1) Method Comparison on RSITMD and RSICD Test Sets: Table I presents the comparative experimental results on the RSITMD and RSICD datasets. The results demonstrate that our model achieves superior performance across multiple metrics. On the RSITMD dataset, our model attains a mean rank (mR) of 42.22, ranking first among current mainstream and state-of-the-art RSITR models. Notably, our model's mean rank surpasses that of the SIRS model (40.21), which is a vision–language model trained on extensive pretraining data. On the RSICD dataset, our model achieves the highest performance in text-to-image retrieval R@10 (41.34), image-to-text retrieval R@5 (31.71), and mean rank mR (28.28). In addition, our model outperforms the MSA model based on contrastive learning CLIP, exceeding its mean rank (22.83).

Furthermore, on both datasets, our model significantly surpasses traditional dual-stream architecture-based image–text retrieval models. On the RSITMD dataset, compared with VSE++ and SCAN, our model achieves improvements of 70% and 58% in mR, respectively. On the RSICD dataset, our model demonstrates improvements of 90% and 99%, respectively. This superior performance is attributed to the fact that these two models were not systematically trained on remote sensing images, while simultaneously demonstrating our model's capability to effectively understand and match relationships between remote sensing images and text. For RSITR models specifically trained on remote sensing images, our model shows a 5% improvement in mR on the RSITMD dataset (42.22 versus 40.21). On the RSICD dataset, our model achieves a 2% improvement in mR (28.28 versus 27.71). These results indicate that our model can perform more precise matching for remote sensing visual images. Due to the more fine-grained nature of text descriptions in RSITMD, our retrieval performance on the RSITMD dataset significantly exceeds that on RSICD.

2) Method Comparison on UCM-Captions Test Set: Following comparative experiments on the RSITMD and RSICD datasets, our model demonstrates superior performance. To validate the model's generalization capability, we conducted additional comparative experimental analysis on the UCM-captions dataset, with results presented in Table II. However, due to the unavailability of code and checkpoints for certain models, we were unable to compare against these methods. As shown in the table, our model achieves excellent performance in text-to-image retrieval R@1 (19.52), image-to-text retrieval R@1 (16.19), and mean rank mR (52.46), surpassing all other models across these metrics. Our model's mean rank improved from 36.93 to 52.46 compared with VSE++. Compared with models specifically trained on remote sensing images, including AMFMN, LW-MCR, SMLGN, MSITA, CDMAN, MSA, and SIRS, our mean rank demonstrates improvements of 14%, 16%, 9%, 4%, 12%, 1%, and 4%, respectively. These results indicate that our model exhibits outstanding performance even on smaller datasets, demonstrating adaptability and strong generalization capability, effectively learning overlooked details in image–text data with limited training samples.

Overall, in RSITR tasks, our model achieves significant improvements in R@1 precision, primarily attributed to the introduction of the OVF method. This approach employs

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TABLE I
METHOD COMPARISON ON RSITMD AND RSICD TEST SETS

Method	RSICD							RSITMD						
	Text Retrieval			Image Retrieval			mR	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [24]	4.56	16.73	22.94	4.37	15.37	25.35	14.89	10.38	27.65	39.60	7.79	24.87	38.87	24.83
SCAN [21]	5.85	12.89	19.84	3.71	16.40	26.73	14.23	11.06	25.88	39.38	9.82	29.38	42.12	26.82
AMFMN [26]	5.39	15.08	23.40	4.90	18.28	31.44	16.42	10.63	24.78	41.81	11.51	34.09	54.87	29.72
LW-MCR [61]	4.39	13.35	20.29	4.30	18.85	32.34	15.59	9.73	26.77	37.81	9.25	34.07	54.03	28.58
GaLR [17]	6.59	19.85	31.04	4.69	19.48	32.13	18.96	14.82	31.64	42.48	11.15	36.68	51.68	31.41
ACF [62]	8.23	20.31	30.47	7.39	23.28	36.91	21.10	15.49	34.96	49.12	12.83	37.74	55.53	34.28
MGRM-EL [63]	7.41	23.24	35.32	5.75	21.23	35.55	21.42	7.41	23.24	35.32	5.75	21.23	35.55	21.42
MSITA [65]	8.67	22.71	33.91	6.13	21.98	35.39	21.47	15.22	34.20	47.65	12.15	39.92	57.72	34.48
CDMAN [66]	10.23	24.94	36.83	7.13	25.38	40.35	24.14	16.46	37.74	51.24	13.82	41.96	59.74	36.83
DOVE [28]	8.66	22.35	34.95	6.04	23.95	40.35	21.72	16.81	36.80	49.93	12.20	44.13	66.50	37.73
MSA [13]	7.50	23.70	36.96	6.77	23.04	39.03	22.83	16.59	40.04	51.33	15.53	44.20	60.80	38.08
SIRS [67]	10.34	27.69	39.66	8.74	30.38	**49.50**	27.71	18.45	44.06	57.73	**19.44**	43.18	58.40	40.21
Ours	**10.52**	**29.43**	**41.34**	9.30	**31.71**	47.36	**28.28**	**20.93**	**44.78**	**59.34**	15.51	**45.40**	**67.35**	**42.22**

TABLE II
METHOD COMPARISON ON UCM-CAPTIONS TEST SET

Method	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [24]	12.38	44.76	65.71	10.10	31.80	56.85	36.93
SCAN [21]	14.29	45.71	67.62	12.76	50.38	77.24	44.67
AMFMN [26]	16.67	45.71	68.57	12.86	53.24	79.43	46.08
LW-MCR [61]	18.10	47.14	63.81	13.14	50.38	79.52	45.25
SMLGN [64]	12.86	49.52	75.71	14.29	52.76	84.67	48.30
MSITA [65]	16.86	49.33	73.33	14.29	57.16	91.58	50.43
CDMAN [66]	10.95	47.43	73.53	12.23	51.83	85.54	46.90
MSA [13]	13.33	**59.05**	77.14	13.14	57.52	**92.48**	52.11
SIRS [67]	15.21	50.89	76.62	14.63	57.31	89.29	50.66
Ours	**19.52**	51.90	**77.62**	**16.19**	**59.52**	90.00	**52.46**

omni-perspective fusion strategies to help the model comprehensively understand detailed associations between images and text, thereby enhancing retrieval accuracy. Through the effective combination of GFA and MFA modules, our model precisely captures both local and global information between images and text, further optimizing image–text matching performance. While demonstrating excellent performance on large-scale datasets, model performance may be constrained by data diversity, particularly on small-scale datasets. For smaller training sets like the UCM-captions dataset, the data may not fully encompass features of all categories, increasing the difficulty of image–text pairing and consequently affecting retrieval accuracy. Nevertheless, despite the limited data scale, our method achieves outstanding performance on the UCM-captions dataset, fully demonstrating RCA-Loss's capability to adaptively separate heterogeneous features and aggregate homogeneous features on small datasets. Through precise capture of small targets, the model maintains efficient and accurate retrieval performance in complex image–text matching tasks.

3) Visualization of Model Attention Regions: In this section, we present attention map visualizations for both baseline models and our proposed model, as illustrated in Fig. 6. The attention visualizations are generated using the GradCAM method [68]. In the heatmaps, red and yellow regions indicate high attention weights, while blue regions represent low attention weights, clearly demonstrating the key regions that models focus on during cross-modal alignment.

Each column in Fig. 6 displays, from left to right: the original remote sensing image, attention visualization results of the baseline model, and attention visualization results of our proposed model, with the corresponding query text shown above each row. From the visualization results, we observe that baseline models tend to focus on global image features or are affected by background noise interference, whereas our model can more precisely localize key semantic elements described in the text.

Specifically, in Fig. 6(a), while the baseline model only captures the global textural features of the desert, our model accurately focuses on the "dry pond" described in the query text. In Fig. 6(b), our model simultaneously attends to both the square church building and the curved road above it, whereas the baseline model's attention distribution is more scattered. In Fig. 6(c), our model precisely focuses on the white terminal building and the aircraft, demonstrating understanding of the semantic mapping between the query text and concrete structures in the airport scene. In Fig. 6(d), the baseline model is distracted by background interference and shows scattered attention, while our model accurately localizes the triangular metal building described in the query text. In Fig. 6(e), our model not only highlights the main viaduct structure but also attends to the sparse trees in the surrounding area, achieving coordinated localization of multiple targets mentioned in the query text.

These results demonstrate that, compared with baseline models, our proposed improved architecture possesses

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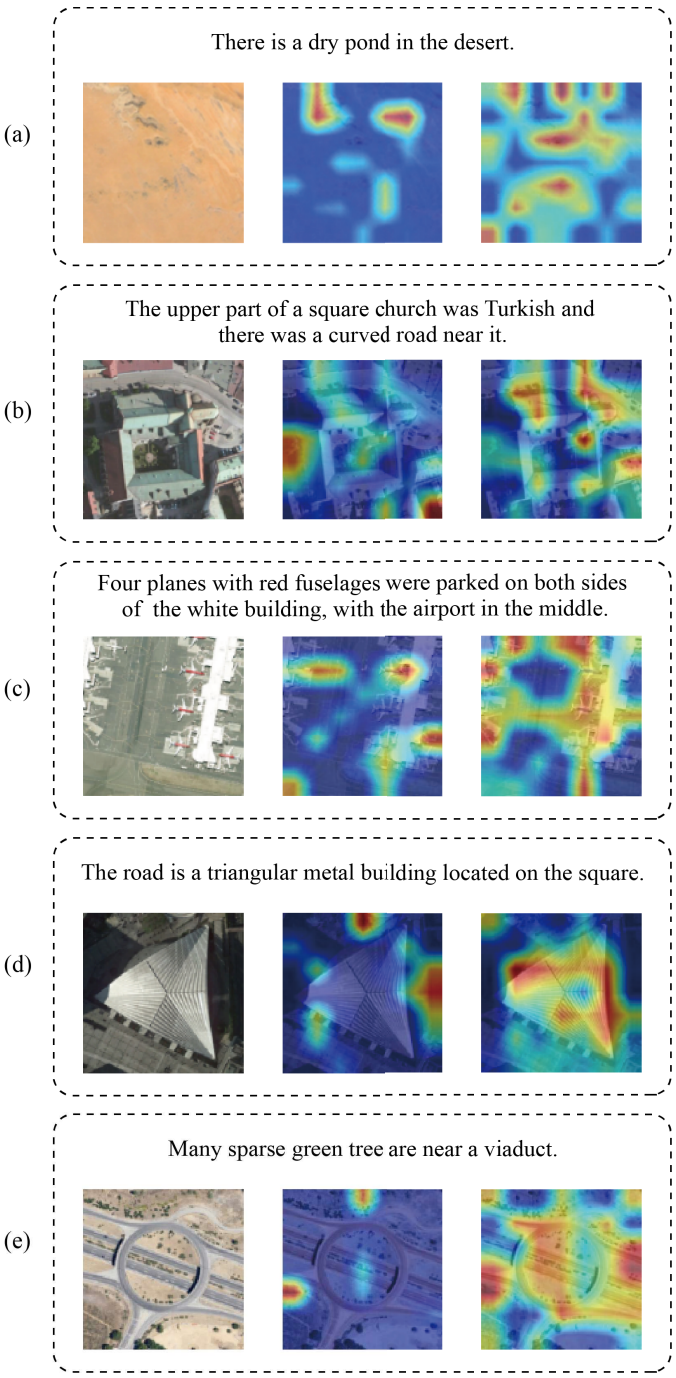


Fig. 6. Visualization of model attention regions for text-to-image retrieval tasks. (Top) Each row shows the query text, (left) original remote sensing image, (middle) baseline model attention visualization, and (right) our proposed model attention visualization. The attention maps demonstrate our model's superior ability to localize semantically relevant regions described in the query text compared with the baseline approach. (a) Desert, (b) church, (c) airport, (d) triangular building, and (e) viaduct.

stronger fine-grained target localization capability, complex scene understanding ability, and noise resistance capability. It can establish more precise semantic correspondences between textual descriptions and image regions, thereby achieving more effective cross-modal feature representation learning.

4) Qualitative Analysis of Bidirectional Retrieval Tasks: In this section, we present qualitative analysis results of the FAME model in bidirectional retrieval tasks. Fig. 7 demonstrates the analysis on the RSITMD dataset, while Fig. 8 shows the analysis on the RSICD dataset. These two figures illustrate that the FAME model exhibits outstanding performance in cross-modal retrieval tasks.

In text-to-image retrieval tasks, the FAME model demonstrates precise retrieval capabilities. On the RSITMD dataset, for three query texts: "there are many white planes parked beside the blue house," "some trees have been planted around the harbor," and "two dark green cracks spread in this yellow desert," the model successfully ranks the correct images in the top position of retrieval results, achieving 100% top-1 accuracy. On the RSICD dataset, the model similarly retrieves images that semantically match the query texts with high accuracy. For the query "there are two ponds of different colors with trees on the bank," the model successfully identifies images containing water bodies of different colors and vegetation along the banks. For the complex spatial relationship description "the bridge over a river and woods along it divides at each bridgehead," the model accurately localizes the corresponding bridge and surrounding environment. For the commercial building description "the diamond commercial is by the roadside near a bareland," the model also precisely matches the target image.

In image-to-text retrieval tasks, the FAME model also exhibits excellent semantic understanding capabilities. On the RSITMD dataset, when given a playground image, the model accurately retrieves descriptions such as "there is a small tennis court with some plants" and "there is a small tennis court surrounded by plants." When faced with an overpass image, the model successfully matches texts that accurately describe the structural features of the overpass, such as "the viaduct has four symmetrical rings of ramps" and "there is a viaduct with four symmetrical loops." For storage tank facility images, the model retrieves detailed descriptions including "three large white storerooms and some green trees are on the bank of the river" and "three huge white storage tanks and some green trees stand by the river." On the RSICD dataset, when given a residential swimming pool image, the model retrieves accurate descriptions such as "rows of palm trees surrounded this smart house and its swimming pool next to this green bush." For overpass images, the model matches descriptions like "the broad viaduct is accessed to the winding road by ramps" and "here is a quite bustling wide flyover spans the curved road below." When faced with coastal building images, the model retrieves precise texts such as "two simple buildings are standing on this beach" and "a piece of green ocean is near a yellow beach with two buildings." Particularly noteworthy is the model's strong adaptability in processing professional terminology and spatial relationship descriptions in remote sensing images, accurately understanding specialized vocabulary such as "symmetrical rings," "viaduct," and "storage tanks."

E. Ablation Studies

To comprehensively demonstrate the impact of different components in our model, we use Swin Transformer_tiny and BERT_base as the baseline and conduct experiments

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Query text	Top-5 retrieval images	Query image	Top-5 retrieval texts
There are many white planes parked beside the blue house.			1.There is a small tennis court with some plants. 2.In most areas, sparse residential areas are covered with green trees. 3.It is a medium residential area . 4.There is a house where the garden where the winding road passes. 5.There is a small tennis court surrounded by.
Some trees have been planted around the harbor.			1.The viaduct has four symmetrical rings of ramps . 2.There is a viaduct with four symmetrical loops. 3.There are three service areas in one corner of the viaduct. 4.There is a viaduct with four symmetrical rings. 5.Roads and viaducts form a cross across the saddle line.
Two dark green cracks spread in this yellow desert.			1.Three large white storerooms and some green trees are on the bank of the river. 2.There are many spherical and cylindrical tanks on the ground. 3.Three huge white storage tanks and some green trees stand by the river. 4.Here are three big white containers and some green trees on the river. 5.Large and small storage spaces are on the ground.

Fig. 7. Qualitative analysis of the FAME model on bidirectional retrieval tasks using three randomly selected remote sensing images and three query texts from the RSITMD test set. Green boxes indicate correct images, and green text indicates correct texts.

Query text	Top-5 retrieval images	Query image	Top-5 retrieval texts
There are two ponds of different colors with trees on the bank.			1.Rows of palm trees surrounded this smart house and its swimming pool next to this green bush. 2.There is a swimming pool behind the brown house. 3.There is a swimming pool behind the house. 4.There is a house near a square coconut grove on the lawn with coconut trees scattered around. 5.A red building with a swimming pool is surrounded by many different green plants.
The bridge over a river and woods along it divides at each bridgehead.			1.The broad viaduct is accessed to the winding road by ramps. 2.Here is a quite bustling wide flyover spans the curved road below. 3.There is a spindle shaped viaduct with many cars. 4.A large number of tall trees are planted on both sides of the road. 5.Many trees and lawns are planted on both sides of the road.
The diamond commercial is by the road side near a bareland.			1.Two simple buildings are standing on this beach. 2.A piece of green ocean is near a yellow beach with two buildings. 3.There is two buildings on the coast. 4.Some flats are on the brown beach near the feculent sea. 5.The coastline is straight and the sea has a fish scale shaped texture.

Fig. 8. Qualitative analysis of the FAME model on bidirectional retrieval tasks using three randomly selected remote sensing images and three query texts from the RSICD test set. Green boxes indicate correct images, and green text indicates correct texts.

on the RSITMD dataset. Table III presents the experimental results. From this table, we can observe that compared with the base module, OVF module achieves a 1.18% improvement in mR, indicating that OVF module can extract omni-perspective information from the global features of Swin-T and BERT, contributing to subsequent feature fusion.

Compared with the OVF module, RCA-Loss demonstrates a more significant improvement of approximately 0.51%, effectively bringing similar features closer while separating dissimilar ones. The GFA module achieves a 1.54% improvement in mR over the baseline, confirming the effectiveness of global feature aggregation. The GFA-MFA module performs importance-based feature selection, ensuring effective extraction of key image–text features. When added to OVF, the GFA-MFA module yields a 1.04% mR improvement. The sixth experimental group reveals that RCA-Loss functions effectively when combined with the OVF module, achieving a 2.57% mR improvement compared with the baseline configuration. Our progressive masking strategy also contributes significantly to model training. Comparing the seventh and eighth experimental groups shows that our complete model outperforms the version without progressive masking by 1.72% in mR.

We further evaluate the computational efficiency of individual modules in our FAME framework, as presented in Table IV. The analysis examines three key metrics: floating-point operations (FLOPs), inference latency, and GPU memory usage. The OVF module demonstrates excellent efficiency

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TABLE III
ABLATION EXPERIMENT FOR DIFFERENT STRUCTURES OF FAME ON RSITMD TEST SET

No	Method	Text Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
1	Base	16.25	41.15	54.20	10.92	40.88	61.78	37.53
2	Base+OVF	17.72	40.42	55.76	12.23	43.80	62.33	38.71
3	Base+GFA	19.25	38.72	53.10	14.65	44.47	64.25	39.07
4	Base+$\mathcal{L}_{\text{RCA}}$	16.38	41.21	57.77	12.78	44.65	62.55	39.22
5	Base+OVF+GFA-MFA	18.38	42.36	56.63	13.70	44.25	63.23	39.75
6	Base+OVF+$\mathcal{L}_{\text{RCA}}$	20.04	43.20	**59.53**	14.91	45.12	64.86	41.28
7	Ours w/o ProgMask	19.12	42.94	56.66	14.13	**45.87**	64.25	40.50
8	Ours	**20.93**	**44.78**	59.34	**15.51**	45.40	**67.35**	**42.22**

TABLE IV
EFFICIENCY EVALUATION OF INDIVIDUAL MODULES IN FAME FRAMEWORK

No	Method	FLOPs (G)	Inference Time (ms)	GPU Memory (MB)
1	Base	10.567	19.58	2457.10
2	Base+OVF	+0.001	+0.58	+3.80
3	Base+GFA	+1.518	+4.33	+293.70
4	Base+OVF+GFA-MFA	+2.651	+7.63	+625.40

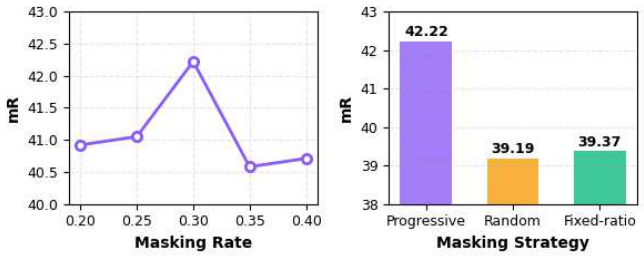


Fig. 9. Ablation study of ProgMask module. (Top) Performance comparison across different masking rates using the progressive strategy. (Bottom) Comparison of masking strategies at masking rate 0.3.

TABLE V
IMPACT OF FILTERING VALUE K IN GFA ON RSITMD TEST SET

k	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
5	17.26	40.27	53.76	13.14	42.88	65.31	38.77
10	16.81	40.49	54.65	14.03	43.94	64.38	39.05
15	18.81	39.82	57.96	12.79	44.78	66.06	40.04
20	15.93	42.26	54.42	13.67	43.58	65.09	39.16
25	**20.93**	**44.78**	**59.34**	**15.51**	45.40	67.35	**42.22**
30	18.83	43.24	57.65	14.12	**46.47**	**67.71**	41.34
35	17.92	41.15	55.75	13.27	44.38	65.49	39.66
40	17.92	40.93	55.09	13.89	45.04	64.12	39.50
45	18.14	42.04	55.53	14.29	44.34	63.67	39.67

with negligible computational overhead, adding only 0.001 GFLOPs, 0.58-ms inference time, and 3.8-MB GPU memory. The GFA module requires more computational resources due to its global feature aggregation mechanism, contributing 1.518 GFLOPs, 4.33-ms inference latency, and 293.7-MB additional GPU memory. When combining both OVF and GFA-MFA modules, the total computational overhead is 2.651 GFLOPs, 7.63-ms inference time, and 625.4-MB GPU memory, representing approximately one-fourth additional cost compared with the base model. Despite this moderate computational cost, our complete framework achieves substantial performance improvements with an absolute gain of 4.69 points in the mR metric (from 37.53 to 42.22).

F. Parameter Analysis

1) Impact of Masking Strategy in ProgMask: To validate the effectiveness of the progressive masking strategy and determine the optimal masking rate, we conducted ablation studies on the ProgMask module using the RSITMD dataset. Fig. 9 presents the experimental results examining both masking rate selection and strategy comparison. The upper subplot shows performance variations across different masking rates using the progressive strategy. The model achieves peak performance at a masking rate of 0.3 with an mR score of 42.22. Lower masking rates, such as 0.25, yield suboptimal results with mR of 41.05, while higher masking rates of 0.35 and 0.4 lead to performance degradation with mR scores of 40.58 and 40.71, respectively.

The lower subplot compares three masking strategies at the optimal rate of 0.3. Progressive masking achieves mR of 42.22, outperforming random masking at 39.19 and fixed-ratio masking at 39.37. Random masking randomly selects masking rates up to 0.3, while fixed-ratio masking consistently applies a masking rate of 0.3. These results confirm that progressive masking provides an optimal balance between robustness enhancement and information preservation.

2) Impact of Filtering Value k in GFA: In the GFA module, the k value controls the number of most relevant features filtered from the score matrix, thereby determining the regions that the model focuses on. To optimize the feature filtering process, we conducted experiments with different k values on the RSITMD dataset to obtain an optimal k value. As shown in Table V, when k equals 25, the model achieves optimal performance, with approximately 9% improvement in mR compared with $k = 5$. When $k < 25$, we observe that model performance gradually degrades, particularly when $k = 5$, where the model performs poorly across all evaluation metrics. This indicates that insufficient feature filtering leads to overly restricted attention regions, preventing the model from

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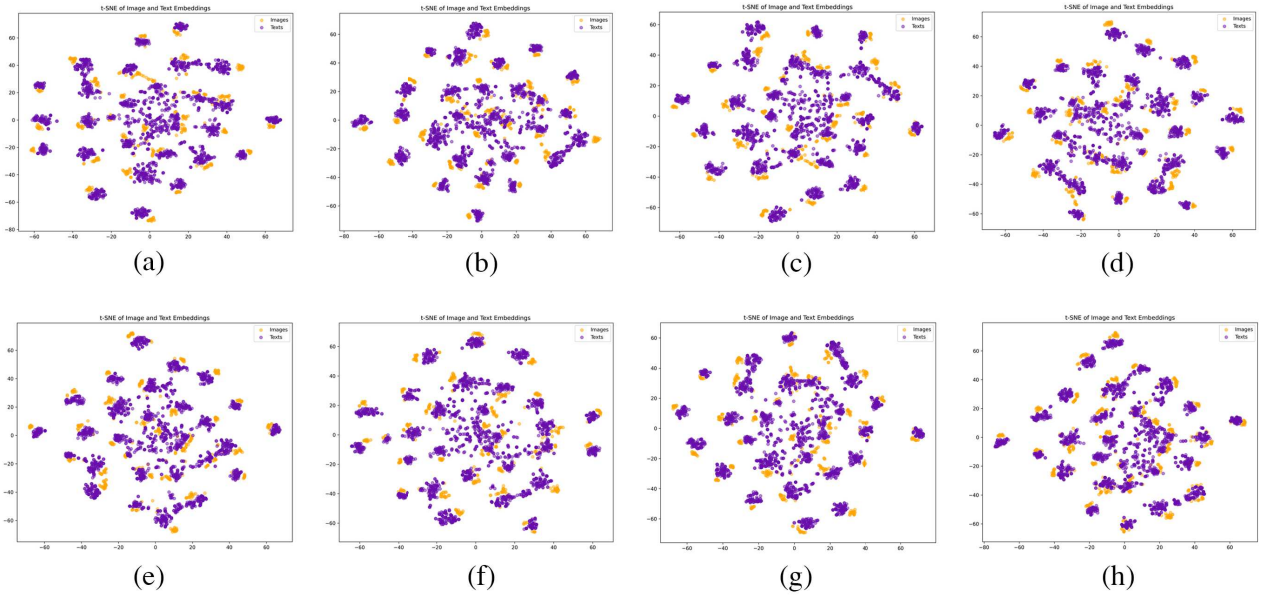


Fig. 10. Visualization of image–text embedding space under different radii on RSITMD test set. (a)–(h) Radius values 0.05, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0, and 2.0, respectively, demonstrating the clustering performance under different RCA-Loss radius settings.

TABLE VI
IMPACT OF RADIUS r IN RCA-LOSS ON RSITMD TEST SET

r	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0.05	16.50	40.65	54.87	14.21	**46.13**	63.72	39.35
0.1	18.32	41.13	55.29	14.45	44.97	64.62	39.80
0.3	17.55	42.26	54.28	13.67	43.85	65.36	39.50
0.5	19.33	**45.28**	57.64	15.27	44.92	66.12	41.43
0.7	**20.93**	44.78	**59.34**	**15.51**	45.40	**67.35**	**42.22**
0.9	18.46	43.95	57.11	14.29	46.12	66.33	41.04
1	17.24	43.32	56.42	14.11	44.37	65.98	40.24
2	16.35	41.76	54.84	13.38	43.92	64.35	39.10

adequately learning the underlying features of the data. When $k > 25$, although model performance continues to improve, the improvement margin gradually diminishes, and all metrics tend to stabilize. This suggests that excessive feature filtering may cause the model to over-focus on certain features, adversely affecting overall performance. Therefore, selecting $k = 25$ as the optimal feature filtering quantity enables the model to balance feature selection and generalization capability, thereby achieving optimal performance.

3) Impact of Radius r in RCA-Loss: During the retrieval process, radius r is a crucial parameter in the RCA-Loss function. It ensures that image and text features from different categories are pulled to distances greater than radius r to prevent feature confusion during feature alignment, while pulling same-category image–text features closer to distances smaller than the radius to enhance the model's sensitivity to category discrimination. Based on the mean distances of different category features, we set r to 0.05, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0, and 2.0 to investigate the impact of different radius values on model performance on the RSITMD dataset. Table VI presents the model's retrieval performance under different radii, while Fig. 10 shows the corresponding image–text embedding space visualization results, where (a)–(h) corresponds to the aforementioned radius values, respectively. Experimental results demonstrate that when $r = 0.7$, the model achieves optimal performance with an mR metric of 42.22, representing an 8% improvement compared with the baseline value (39.1 when $r = 2.0$). From the quantitative results, we observe that when the radius is smaller than 0.5, the mR metric is relatively low (39.46–39.80). When the radius is set to 0.5–0.9, performance significantly improves, while performance declines after exceeding 1.0. This trend is well validated in the visualization results of Fig. 10(e): when $r = 0.7$, data points from different categories form the clearest clustering boundaries, with high intraclass cohesion and large interclass separation. When the radius is too small, clustering boundaries become blurred, while when the radius is too large, it leads to loose clustering structures. It should be noted that although some visualization results in Fig. 10 show minimal visual differences, this is due to information loss during the t-SNE dimensionality reduction process and limitations in mapping from high-dimensional feature space to 2-D space. The differences in quantitative metrics more accurately reflect the actual performance changes of the model in the original high-dimensional feature space. These results indicate that an appropriate radius value can maintain tight clustering of same-category samples while effectively avoiding feature confusion between different categories. Too small a radius leads to insufficient constraints, while too large a radius causes over-constraint and affects model performance.

4) Impact of GFA-MFA Layer Number N: This section compares the performance of the GFA-MFA module under different layer numbers, as shown in Table VII. When the layer number N of the GFA-MFA module is less than 3, model performance significantly improves with increasing layer numbers, indicating that the increase in layers enhances the comprehensiveness and granularity of feature fusion. When $N = 3$, the model's mR metric reaches its optimal value. However, when $N > 3$, model performance begins to decline,

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TABLE VII
IMPACT OF GFA-MFA LAYER NUMBER N ON RSITMD TEST SET

N	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
2	17.04	43.58	55.97	11.46	45.04	63.54	39.44
3	**20.93**	**44.78**	59.34	**15.51**	**45.4**	**67.35**	**42.22**
4	19.47	41.81	**60.52**	14.98	44.98	65.69	41.24
5	18.14	38.72	53.10	10.97	42.61	64.47	38.00
6	18.36	40.04	55.53	13.05	45.31	66.24	39.76
7	18.14	42.04	55.09	14.42	44.12	65.18	39.83
8	19.25	41.59	54.42	14.56	42.57	63.72	39.35

suggesting that excessively deep layer structures do not further enhance model effectiveness and may instead lead to overfitting.

V. CONCLUSION

This article proposes the FAME model, which integrates four core modules: ProgMask, OVF, GFA-MFA, and RCA-Loss to enhance cross-modal alignment in RSITR. To accurately capture remote sensing geographical image–text objects, we employ the OVF module to comprehensively extract local features of images and text through OVF, ensuring fine-grained information preservation. Simultaneously, the GFA-MFA module precisely captures and fuses fine-grained features of images and text through global feature filtering and multiperspective feature alignment. The combination of these two modules establishes robust correlations between local and global features, thereby enhancing the synergistic effects of cross-modal features. RCA-Loss further strengthens the accuracy and robustness of image–text alignment by optimizing the aggregation of same-category features and the separation of different-category features. In addition, we employ ProgMask to guide model learning. We conducted evaluations on three public remote sensing image caption datasets (RSITMD, RSICD, and UCM-captions), and the results demonstrate that our method achieves superior performance compared with several existing state-of-the-art methods across all three datasets. Meanwhile, ablation experiments validate the effectiveness of OVF, GFA-MFA, RCA-Loss, and ProgMask.

While our method demonstrates strong performance in RSITR tasks, numerous potential directions remain for further exploration and improvement. Currently, many pretrained vision–language foundation models have been widely applied to multimodal tasks. Future research directions include integrating these foundation models into RSITR tasks with specifically designed fusion strategies to better adapt to remote sensing applications, thereby enhancing model performance. In addition, regarding parameter optimization, model compression techniques can be explored to improve computational and storage efficiency while maintaining high-precision retrieval.

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